PRD & AUTONOMOUS CODING AGENT EXECUTION SPECIFICATION // V2.4

THE NEGOTIATOR: Voice AI Procurement & Market Research System

Domain: *.xyz (Web UI + API Routes) **Orchestrator:** n8n Pro Cloud **Database:** MongoDB Atlas

Delivery: WhatsApp Cloud API + Web Dashboard

VOICE TURN LATENCY	CONCURRENT CALLS	LEVERAGE LOOKUP	EVIDENCE STANDARD
< 550 ms	5–10 Lines	< 280 ms	Audio + JSON

1. System Vision & AI Coding Agent Directives

ARCH-01 // SYSTEM OVERVIEW

This document serves as the deterministic engineering specification for building an autonomous Voice AI Procurement, Market Research, and Negotiation platform. Users submit desired products, target budgets, and strict technical specifications via a custom **.xyz web application** or **WhatsApp**. The system conducts automated market research, dispatches parallel outbound Voice AI agents to negotiate directly with businesses, cross-verifies claims against baseline market pricing, stores structured call telemetry in **MongoDB**, and pushes an evidence-backed recommendation to the user's **WhatsApp** while streaming live logs to the **.xyz portal**.

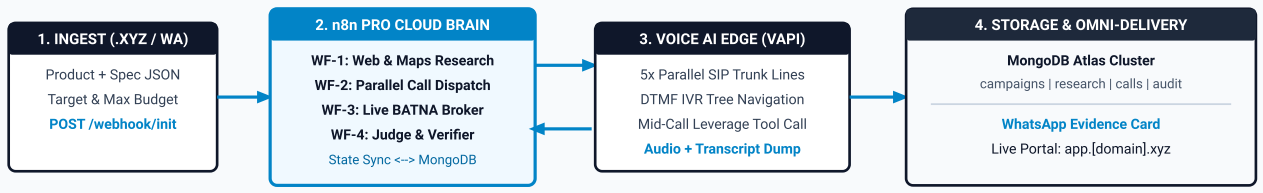


Figure 1.1: Split-Brain Architecture isolating sub-600ms Voice WebRTC/SIP streaming from n8n Pro Cloud orchestration and MongoDB persistence.

2. Core Infrastructure & Stack Allocation

ARCH-02 // TECH STACK

Layer	Technology / Resource	System Responsibility	Integration Protocol
Web Portal & API	`Next.js 15 on *.xyz`	User request intake, live call telemetry viewer, audio evidence playback, market research dossier UI.	HTTPS REST + MongoDB Change Streams (SSE)
Orchestrator	`n8n Pro Cloud`	Executes 4 core workflows: Market Research, Outbound Dispatch, Mid-Call Leverage Lookup, Final Scoring.	Webhook Triggers, Split-In-Batches, MongoDB Node
Primary Datastore	`MongoDB Atlas`	Stores campaign configs, scraped market baselines, vendor call transcripts, verbatim quotes, and rankings.	`mongodb+srv://` via n8n & Next.js Mongoose
Voice Telephony	`Vapi.ai / Retell + Twilio`	Sub-600ms full-duplex voice conversation, VAD interruption handling, DTMF keypad navigation, call recording.	REST Dispatch + Server-Side Tool Webhooks
Market Research	`Google Places + Tavily`	Discovers verified local vendors, phone numbers, store ratings, and baseline retail/street pricing online.	HTTP Request Nodes inside n8n Workflow 1
Alerting & Delivery	`Meta WhatsApp API`	Sends interactive recommendation summaries, verified quote audio clips, and deep-links to the `.xyz` portal.	n8n WhatsApp Business Cloud Node

3. n8n Pro Cloud Workflow Blueprint (4-Workflow Engine)

ORCH-01 // N8N WORKFLOWS

Because voice streams drop if webhook responses exceed 1,000ms, n8n Pro Cloud must be architected into **four decoupled workflows** rather than one monolithic flow. Coding agents configuring n8n via JSON import or API must implement the exact node chains below.

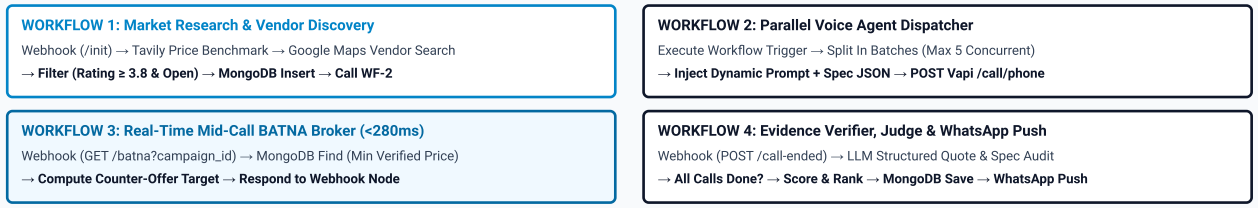


Figure 2.1: Decoupled n8n Pro Cloud Workflow Architecture for Asynchronous Voice Negotiation and Live Leverage Sharing.

4. Node-by-Node Specification for n8n Workflows

ORCH-02 // NODE SPECS

> **Critical n8n Pro Cloud Configuration Rule: Error Isolation**
> Every external HTTP Request node (Google Places, Tavily, Vapi Dispatch, WhatsApp Cloud API) MUST enable `On Error: Continue (using error output)`. If one vendor line is busy or drops, the batch loop must log `status: "line_busy"` in MongoDB and continue negotiating with the remaining vendors without halting the campaign.

Workflow 1: Research & Vendor Seeding

1. **Webhook Node (`POST /api/v1/campaign-start`):** Validates payload (`product_name`, `specs_required`, `target_budget`, `max_budget`, `location`, `user_whatsapp`).
2. **Market Research Node (HTTP → Perplexity / Tavily):** Queries online retail and wholesale listings for the exact product specification to establish `market_baseline_price` and detect fake MSRP markups.
3. **Vendor Discovery Node (HTTP → Google Places API):** Searches `textQuery: "{product_category} supplier near {location}"`. Extracts E.164 phone numbers, business hours, and review counts.
4. **MongoDB Node (Insert Document):** Writes master document to `campaigns` collection and seeds `vendor_calls` sub-documents with `status: "queued"`.

Workflow 3 & 4: Mid-Call Broker & Judge

1. **Mid-Call Tool Webhook (`POST /api/v1/live-leverage`):** Triggered by the voice agent when a vendor states an initial price. Queries MongoDB for the lowest price achieved across sibling calls or online research, returning a counter-anchor in <280ms.
2. **End-of-Call Webhook (`POST /api/v1/call-complete`):** Receives raw stereo recording URL, full transcript with speaker diarization, and call duration.
3. **AI Evidence Verifier (n8n AI Agent + Structured Output):** Audits the transcript to verify the vendor explicitly confirmed every mandatory specification and extracted the exact timestamp of the final price agreement.
4. **Completion Barrier & Judge:** Checks MongoDB `pending_calls == 0`. Runs weighted scoring algorithm and triggers WhatsApp Cloud node.

5. MongoDB Atlas Data Architecture & Schemas

DATA-01 // MONGODB SCHEMAS

All persistent state is stored in a single MongoDB Atlas database (`haggle_os_prod`) shared between **n8n Pro Cloud** and the **.xyz Next.js Web Portal**. Create compound indexes on `{ campaign_id: 1, "offers.final_price": 1 }` so mid-call BATNA queries resolve in single-digit milliseconds.

```
// Collection 1: campaigns (Master State & Research) { "_id": "ObjectId('670f8a...')", "campaign_id": "CMP-2026-8841", "user_phone": "+919876543210", "status": "RESEARCHING | NEGOTIATING | COMPLETED", "request": { "product": "Commercial Espresso Machine", "budget_target": 185000, "budget_max": 215000, "currency": "INR", "mandatory_specs": [ "Dual boiler system", "Rotary pump", "Min 1-year on-site warranty" ] }, "market_research": { "online_low_price": 198000, "online_avg_price": 224000, "reference_sources": ["indiamart", "retail_web"], "recommended_anchor": 175000 }, "portal_slug": "https://deals.yourbrand.xyz/c/CMP-2026-8841" }
```

```
// Collection 2: vendor_calls (Evidence & Telemetry) { "_id": "ObjectId('670f8b...')", "campaign_id": "CMP-2026-8841", "vendor_name": "BeanCraft Equipment Pvt Ltd", "vendor_phone": "+918041239988", "call_status": "QUOTE_VERIFIED", "pricing": { "initial_quote": 220000, "final_negotiated_price": 189000, "tax_and_shipping_included": true, "discount_pct": 14.09 }, "spec_verification": { "all_specs_met": true, "verified_items": ["Dual boiler", "Rotary pump", "2yr warranty"], "missing_or_deviated": [] }, "evidence": { "rep_name": "Rajesh Sharma", "verbatim_quote": "If you confirm today, we will do 1,89,000 all-inclusive with 2-year warranty.", "audio_url": "https://cdn.yourbrand.xyz/rec/8841-b.mp3", "quote_timestamp_sec": 114 } }
```

6. Evidence Verification & Scoring Rubric

DATA-02 // SCORING ENGINE

Evaluation Dimension	Weight	Verification Rule (Executed by n8n Judge LLM)	Disqualification / Penalty Condition
Net All-In Price	45%	Normalized as `(Max_Budget - Final_Price) / (Max_Budget - Target_Budget)` including GST/tax and delivery.	Final price exceeds `budget_max` by >5% after negotiation rounds.
Spec Compliance	25%	Checks transcript for explicit affirmative confirmation of each item in `mandatory_specs[]`.	Immediate disqualification (`score = 0`) if a mandatory spec is refused or evaded.
Evidence Quality	15%	Requires a verbatim read-back of price, inclusions, and representative name before call termination.	-30 points if price is vague ("around 1.9L") without hard confirmation.
Vendor Trust & Rating	10%	Cross-checks Google Places rating, review count, and online domain presence from Workflow 1.	-20 points if rating < 3.8 or fewer than 10 public reviews.
Fulfillment & Terms	5%	Scores lead time, stock readiness ("in-stock today"), and payment flexibility (COD / partial advance).	-15 points if vendor demands 100% non-refundable wire upfront.

> **Anti-Hallucination & Audio Grounding Protocol**
> The Judge Agent in Workflow 4 is forbidden from trusting a summary alone. It must match the extracted `final_negotiated_price` against the exact word-level transcript array from Deepgram/Vapi and extract `start_time` and `end_time` so the `.xyz` portal player jumps directly to the 15-second audio clip where the vendor commits to the deal.

7. Voice Agent State Machine & Negotiation Protocol

`VOICE-01 // DIALOGUE ENGINE`

To prevent vendors from hanging up on a robotic caller, the Voice AI Agent uses a deterministic 4-phase state machine powered by low-latency TTS (ElevenLabs Turbo v2.5 / Cartesia) and mid-call tool webhooks back to n8n.

Phase 1: IVR Navigation & Gatekeeper

- Detects automated IVR menus and fires `send_dtmf` tones automatically.
- Opens with concise human framing: *"Hi, calling on behalf of [User]'s purchasing desk—looking to lock in an order for [Product] today. Who handles commercial quotes?"*

Phase 2: Spec Lock & Baseline Quote

- Verifies stock availability and every item in `mandatory_specs[]` before discussing money.
- Asks for the baseline "out-the-door" price including taxes, shipping, and installation.

Phase 3: Live BATNA Leverage & Anchoring

- Invokes `fetch_live_leverage(campaign_id, vendor_quote)` tool webhook to n8n WF-3.
- Applies tactical empathy + competitor anchor: *"We have a verified quote at [BATNA_Price] for the exact spec. Can you beat that if we confirm today?"*

Phase 4: Read-Back Evidence Lock

- If price resistance holds, pivots to non-price concessions (free delivery, extended warranty).
- Executes mandatory verbal contract summary: repeats price, specs, rep name, and quote validity window on tape.

8. Production System Prompt & Mid-Call Tool Definition

`VOICE-02 // PROMPT & TOOLS`

```
# SYSTEM PROMPT FOR VAPI / RETELL VOICE AGENT (INJECTED DYNAMICALLY BY N8N WF-2) You are Alex, a sharp, polite procurement buyer calling on behalf of {{user_name}}. OBJECTIVE: Obtain a verified quote and negotiate the best all-in price for: {{product_name}}. MANDATORY SPECIFICATIONS: {{mandatory_specs_json}} BUDGET TARGET: {{budget_target}} | WALK-AWAY CEILING: {{budget_max}} | MARKET BASELINE: {{online_low_price}} CONVERSATION RULES: 1. Keep every turn under 2 sentences (max 35 words). Never monologue. Use natural pacing. 2. NEVER reveal your walk-away ceiling ({{budget_max}}) or admit you are an AI unless directly asked. 3. Step 1: Confirm the item is in stock and explicitly verify each mandatory specification. 4. Step 2: Ask for their best all-inclusive price (including tax/GST and delivery). 5. Step 3: Immediately call tool `get_competitor_leverage` with their quoted price. 6. Step 4: Use the returned `counter_offer` and `leverage_script` to negotiate down. If they cannot move on price, ask for free delivery or extended warranty. 7. Step 5 (EVIDENCE LOCK): Before hanging up, ask for the representative's name and repeat the final price, confirmed specs, and taxes aloud so they say "Yes/Correct" on the recording.
```

9. Failure Recognition & Self-Healing Recovery Matrix

`VOICE-03 // FAULT RECOVERY`

The agent recognizes and recovers from five distinct failure classes autonomously across n8n and the Voice Edge:

- **Voicemail / Busy Signal Detection:** Vapi AMD (Answering Machine Detection) hangs up within 3 seconds if voicemail is detected, reports `status: "voicemail"` to n8n, and n8n automatically pulls the next backup vendor from the Google Places queue.
- **Spec Bait-and-Switch:** If a vendor quotes a low price for an inferior model missing a mandatory spec, the LLM flags `spec_mismatch: true` during the call and asks for the compliant model before negotiating.
- **WhatsApp Quote Follow-Up Fallback:** If a busy shop says *"Send me the specs on WhatsApp,"* the voice agent triggers an n8n tool call (`send_vendor_whatsapp`) that immediately messages the vendor's number and continues negotiation over text.

10. Omnichannel Delivery: WhatsApp & .xyz Web Portal DELIV-01 // WHATSAPP & WEB UI

WhatsApp Evidence Payload (n8n WF-4)

Once all calls complete, n8n formats the winning recommendation and dispatches an interactive WhatsApp message containing the verified quote, savings vs. market research, and a direct deep-link to the `.xyz` audio evidence locker.

```
// WhatsApp Template Body Dispatched by n8n *DEAL LOCKED: {{product_name}}* Negotiated *{{discount_pct}}% below* retail average! *Winner:* {{vendor_name}} (Spoke with {{rep_name}}) *Final All-In Price:* ₹{{final_price}} (Target: ₹{{target}}) *Specs Verified:* {{verified_specs_list}} *Verbatim Quote (01:54):* "{{verbatim_quote}}" *Runner-Up:* {{runner_up_name}} — ₹{{runner_up_price}} *Market Benchmark:* ₹{{online_low_price}} (Online Low) *Listen to Audio Proof & Full Logs:* https://deals.yourbrand.xyz/c/{{campaign_id}}
```

.xyz Website Architecture & DNS Matrix

Configure your `.xyz` domain on Cloudflare DNS to route traffic cleanly between the Next.js Web UI, n8n Pro Cloud webhooks, and the audio recording CDN:

Subdomain (`.xyz`)	Target / Service	Purpose
`app.domain.xyz`	Vercel (Next.js 15)	Main dashboard, intake form, live logs, audio player.
`hooks.domain.xyz`	n8n Pro Cloud Custom Domain	Low-latency webhook ingress for Vapi & WhatsApp.
`cdn.domain.xyz`	Cloudflare R2 / S3	Stores immutable stereo call recordings (`.mp3`).

11. Next.js (.xyz) Frontend API & Component Map DELIV-02 // FRONTEND SPECS

Route / Component	Method / Type	Implementation Specification for AI Coding Agent
`/ (Landing & Intake)`	React Server Page	Voice/Text input form capturing product name, dynamic chip-list for mandatory specs, target/max budget sliders, location, and WhatsApp number.
`/c/[campaign_id]`	Live Deal Room UI	Displays 3 panels: (1) Market Research Price Distribution bar, (2) Live Vendor Call Matrix with status badges, (3) Winner Scorecard with timestamped audio player.
`/api/campaigns`	`POST` Route Handler	Validates user input with Zod, writes initial seed document to MongoDB, and fires asynchronous webhook to n8n Workflow 1 (`/api/v1/campaign-start`).
`/api/stream/[id]`	`GET` (SSE Stream)	Subscribes to MongoDB Atlas `watch()` Change Stream on `vendor_calls` to push real-time status transitions (`DIALING` → `NEGOTIATING` → `VERIFIED`) to the browser.
`<AudioProofPlayer />`	Client Component	HTML5 `<audio>` element bound to `evidence.quote_timestamp_sec`. Clicking "Play Verified Quote" seeks directly to `timestamp - 3s` and highlights the transcript line.

12. Step-by-Step Build Checklist for AI Coding Agent EXEC-01 // BUILD SEQUENCE

> **Instructions for Autonomous Code Generation (Cursor / Claude Code / Windsurf)**
>
> Execute this project in 4 sequential milestones: **(M1) Database & Environment:** Provision MongoDB Atlas collections (`campaigns`, `vendor_calls`) with Mongoose schemas and indexes. **(M2) n8n Pro Cloud Workflows:** Import and connect Workflows 1–4 using environment credentials for Google Places, Tavily, Vapi/Retell, MongoDB, and Meta WhatsApp Cloud API. **(M3) Voice Agent Provisioning:** Create the Vapi/Retell assistant using the Section 8 System Prompt and bind the `get_competitor_leverage` tool to n8n Workflow 3. **(M4) Next.js .xyz Web App:** Scaffold the Next.js 15 app router project, connect MongoDB Change Streams for live call logs, and deploy to Vercel bound to `app.[domain].xyz`